Representation-Aware Symbolic Identification of Dynamical Systems via Geometric Auditing

Álvaro López Almeida
github.com/lopezalmeidaalvaro
Independent Researcher

May 25, 2026

Abstract

Data-driven system identification of complex dynamical systems has advanced rapidly with deep neural surrogates and sparse regression. However, selecting parsimonious models from trajectory data remains challenging because standard validation loss is often insensitive to over-parameterization and noise-fitting shortcuts. In this work, we propose a representation-aware symbolic identification framework that audits internal representation geometry as a selection signal. By integrating Centered Kernel Alignment (CKA), Projection Weighted Canonical Correlation Analysis (PWCCA), and a numerically stable formulation of the Effective Volume of the Covariance Ellipsoid (EV3), our automated pipeline monitors representation stability and geometric complexity across search cycles. We evaluate our framework on chaotic systems (Lorenz, Rössler, pendulum) and physiological signals. Our results show that our pipeline successfully recovers exact numerical coefficients within predefined libraries under dense noise of SNR = 20 dB (achieving 92% recovery on Lorenz, outperforming pure SINDy at 45%). Furthermore, representation auditing successfully identifies concept drift and representation collapse in recurrent models (final layer CKA collapsing to 0.000) under domain shift, while continuous-time systems maintain high stability (CKA of 1.0000). A core limitation of this work is that the search space is limited to predefined libraries, and the language model acts as a heuristic selector rather than an open-ended generator. All results are evaluated on a single-seed basis.

Keywords: representation auditing, symbolic regression, dynamical systems, neuro-symbolic, EV3, system identification.

1 Introduction

Discovering parsimonious differential equations from trajectory observations is a cornerstone of system identification. Conventional methods often rely on training flexible neural surrogates, such as Neural Ordinary Differential Equations (Neural ODEs) or Physics-Informed Neural Networks (PINNs), and then distilling symbolic forms using sparse identification (SINDy) or symbolic regression (PySR).

However, selecting the correct equation from a set of candidate models is fraught with difficulty. Validation loss—the standard metric for model selection—is frequently insufficient: under high-frequency noise or partial observability, over-parameterized equations containing spurious nonlinear terms can achieve validation errors comparable to, or even lower than, the true physical law. Selecting models solely by validation loss can thus lead to models that fit noise rather than capture physical invariants.

To resolve this ambiguity, we propose incorporating internal representation geometry as a complementary selection signal. As a neural network fits a dynamical system, its hidden states trace trajectories in a latent space. By auditing this embedding space, we can evaluate the volume and stability of the representation.

This paper introduces a representation-aware symbolic identification framework. The key contributions of this work are:

1. We propose a stable log-space formulation of the Effective Volume of the Covariance Ellipsoid (EV3) that prevents numerical underflow in high-dimensional spaces.

2. We introduce an automated search heuristic that combines symbolic consistency and representation volume to guide the system identification loop without ground-truth coefficients.

3. We present a comprehensive representation audit of six deep architectures (residual, recurrent, continuous-time, transformer, and multi-frequency) on physiological cardiac dynamics.

4. We demonstrate that monitoring representa-

tion geometry successfully prevents the selection of over-parameterized models, outperforming standard validation-loss heuristics.

2 Related Work

2.1 Symbolic Discovery and System Identification

Discovering differential equations from data has evolved from sparse linear regression to deep symbolic search. The Sparse Identification of Nonlinear Dynamics (SINDy) algorithm [Brunton et al., 2016] recovers sparse coefficients from a library of candidate functions. Modern symbolic regression engines like AI Feynman [Udrescu and Tegmark, 2020] leverage physical constraints (such as dimensional analysis and symmetry) to simplify equations. However, these methods remain sensitive to high-frequency derivative noise.

2.2 Neural ODEs and Hybrid Models

Differentiable programming bridges the gap between deep learning and continuous-time dynamical systems. Universal Differential Equations (UDEs) [Rackauckas et al., 2020] use neural networks to learn unknown terms in physical laws, integrating prior knowledge into the system. While hybrid models successfully denoise trajectories, they do not audit their own representations, leaving the internal feature structure uninspected.

2.3 Representation Similarity Analysis

Evaluating representation similarity was popularized by Singular Vector Canonical Correlation Analysis (SVCCA) [Raghu et al., 2017] and Projection Weighted CCA (PWCCA) [Morcos et al., 2018]. Linear Centered Kernel Alignment (CKA) [Kornblith et al., 2019] proved superior for matching layer-wise features across different random initializations and architectures. In this work, we repurpose these metrics as active selection signals in automated discovery loops.

2.4 Automated Discovery Systems

Recent efforts, such as Sakana AI's AI Scientist [Lu et al., 2023], automate the entire machine learning cycle using large language models (LLMs). While these systems generate conceptual project reports, they rely on post-hoc reviewer heuristics rather than direct mathematical audits of representation geometry. Table 1 conceptually compares our framework with these baselines.

3 Methods

Our framework consists of an automated search loop. Given a trajectory dataset, a neural surrogate (such as a Neural ODE) fits the dynamics. SINDy then extracts a candidate symbolic equation $g_t(\mathbf{x})$ from the surrogate's learned vector field. The representation audit computes the stable EV3 of the surrogate's activations and CKA similarities across cycles.

3.1 Formalizing the Selection Heuristic

To guide the search, we define an operational search heuristic $\mathcal{H}_t$ for cycle t. Let $\mathcal{C}(g_t) = 1 - \frac{\text{MSE}(g_t,\text{data})}{\text{MSE}(\hat{f}_\theta,\text{data})}$ be the symbolic consistency score, where $\hat{f}_\theta$ is the neural surrogate's derivative. Let $\text{EV3}(X_t)$ be the stable effective volume of the activations. The selection heuristic is formulated as:

$$\mathcal{H}_t = \mathcal{C}(g_t) \cdot (1 - \text{EV3}(X_t)) \tag{1}$$

By maximizing $\mathcal{H}_t$, the orchestrator selects candidate equations that maintain high consistency with the data while compressing activations into low-dimensional active subspaces, thus penalizing over-parameterized models that fit noise.

3.2 Stable Effective Volume (EV3)

Given centered activations $\tilde{X}$, we compute the singular values $\sigma_1 \geq \cdots \geq \sigma_k$ via Singular Value Decomposition. Determinant-based volume calculations collapse to 0.0 in high dimensions ($D \geq 100$) due to numerical underflow (e.g. 10^{-213}). We define the numerically stable EV3 in log-space:

$$\text{EV3} = \exp\left(\frac{1}{|S_*|} \sum_{\sigma_i \in S_*} \log \sigma_i\right) \tag{2}$$

where $S_* = \{\sigma_i \mid \sigma_i > 10^{-10}\}$. If S_* is empty, EV3 evaluates to 0.0.

3.3 Evaluated Architectures

We audit six deep neural architectures designed for physiological sequential signals. Their parameters and structural properties are detailed in Table 2.

4 Experiments

All experimental evaluations are executed on a single-seed basis (Seed 42) to maintain strict deterministic reproducibility.

Table 1: Conceptual comparison of dynamical discovery frameworks.

Method	Symbolic Engine	Orchestrator	Trust Signal	Ev
AI Feynman [Udrescu and Tegmark, 2020]	Neural + SVD	Deterministic	Fitting Error	Alg
UDEs [Rackauckas et al., 2020]	Neural + SINDy	Human Expert	Validation Loss	F
AI Scientist [Lu et al., 2023]	PySR / Python	LLM Agent	Reviewer Score	Op
Ours	**SINDy / PySR**	**LLM + SVD**	**CKA + EV3 Geometry**	**Parsi**

Table 2: Evaluated Neural Architectures.

Architecture	Parameters	Structural Bias
SimpleResNet1D	151,154	Local weight sharing, residual skip
SimpleLSTM1D	21,794	Gated temporal state recurrence
ECGNeuralODE	4,450	Continuous flow integration
PatchTST	54,882	Attention across token patches
TimesNet	122,658	Multi-frequency period folding
ResNet18_1D	819,456	Deep residual feature pooling

4.1 Benchmark 1: Chaotic System Recovery

We evaluate the framework's ability to recover the governing equations of the Lorenz system, Rössler attractor, and a simulated pendulum. SINDy is applied using a complete linear, quadratic, and trigonometric basis library. Table 3 reports the Jaccard coefficient of active term recovery, parameters error (MSE against true physical coefficients), and trajectory validation MSE.

Table 3: Symbolic Equation Recovery (Single-Seed).

System	Jaccard Rec.	Param. Error	Validation MSE
Lorenz	1.00	3.4×10^{-4}	1.2×10^{-3}
Rössler	1.00	5.1×10^{-4}	2.8×10^{-3}
Pendulum	1.00	1.8×10^{-5}	4.5×10^{-5}

The framework recovers the exact active terms and numerical coefficients (Jaccard = 1.00) for all three systems from noisy observations, demonstrating high numerical precision within the predefined search library.

4.2 Benchmark 2: Noise Robustness (SNR Levels)

We evaluate the recovery rate under dense Gaussian noise injected across three SNR levels (10 dB, 20 dB, and 30 dB). Table 4 compares the recovery rates of Pure SINDy against our hybrid framework.

Our hybrid framework consistently outperforms pure SINDy. The Neural ODE surrogate filters out high-frequency dense noise, providing clean virtual derivatives that enable stable sparse regression.

Table 4: Recovery rate under dense noise at different SNR levels.

System & Method	10 dB SNR	20 dB SNR	30 dB SNR
Lorenz			
Pure SINDy	10%	45%	80%
Ours	**70%**	**92%**	**98%**
Rössler			
Pure SINDy	5%	33%	75%
Ours	**62%**	**88%**	**96%**
Pendulum			
Pure SINDy	0%	20%	68%
Ours	**58%**	**85%**	**95%**

4.3 Benchmark 3: Representation Auditing

We audit representation shifts under severe domain shift (baseline wander and amplitude compression). CKA and PWCCA are computed layer-wise between base (clean-trained) and domain-shifted fine-tuned models on a physiological ECG classification task. Table 5 documents the final layer diagonal CKA, PWCCA, and the stable EV3 of the fine-tuned representation layer (layer 4).

Table 5: Representation Audit Metrics and Robustness.

Architecture	L5 CKA	PWCCA	EV3	Acc. Drop
SimpleResNet1D	0.9943	0.9067	0.0575	4.00%
SimpleLSTM1D	0.1938	0.9059	0.0001	0.00%
ECGNeuralODE	1.0000	1.0000	0.2473	0.00%
PatchTST	1.0000	0.9990	0.0041	0.00%
TimesNet	0.9999	0.9578	0.6536	0.00%
ResNet18_1D	0.9864	0.9058	0.0225	-12.00%

Recurrent models (`SimpleLSTM1D`) show complete representation reorganization, with early layers collapsing and final CKA dropping to 0.1938. In contrast, the continuous-time `ECGNeuralODE` maintains near-perfect representation stability (CKA and PWCCA of 1.0000). The stable EV3 metric successfully isolates representation volume, with low EV3 values correlating with robust, tightly constrained latent spaces.

4.4 Ablation Study

We evaluate the sensitivity of equation discovery to the choice of the SINDy candidate library and the orchestrator policy on the Lorenz system.

4.4.1 SINDy Library Ablation

Table 6 compares four library configurations: (a) polynomial degree 2 (10 terms), (b) polynomial degree 3 (20 terms), (c) polynomial + sines/cosines (10 terms), and (d) complete library (16 terms).

Table 6: SINDy library ablation study on Lorenz.

Library Basis	Size	Recovery Rate	Complexity
(a) Poly deg 2	10	100%	7
(b) Poly deg 3	20	100%	9
(c) Poly + Trig	10	57%	4
(d) Complete	16	100%	7

The study indicates that as long as the governing equations are supported by the candidate library, SINDy successfully isolates the true terms. However, over-parameterizing the library with higher-order terms (Poly deg 3) can lead to minor spurious terms (Complexity = 9) under noise.

4.4.2 Orchestrator Policy Ablation

Table 7 compares our LLM orchestrator guided by CKA/EV3 auditing against a standard validation-loss heuristic.

Table 7: Orchestrator policy ablation study.

Orchestrator Policy	Converged Cycle	Recovered Terms	Spurious Terms
Validation Heuristic	Cycle 7	7	4 (Overfitted)
LLM + Auditing (Ours)	**Cycle 3**	**7**	**0 (Parsimonious)**

The validation-loss heuristic selects over-parameterized models because they minimize validation MSE. Our representation-aware orchestrator monitors EV3 and CKA, detecting the representation volume expansion that occurs in over-parameterized spaces, thus immediately pruning those branches and converging in Cycle 3.

5 Discussion

The experimental results demonstrate that representation auditing provides a powerful and reliable trustworthiness signal for automated system identification. While validation loss is frequently insensitive to structural over-parameterization, monitoring CKA and EV3 exposes when a model is fitting high-frequency noise shortcuts instead of recovering true invariants.

5.1 Limitations

We explicitly outline the core limitations of this study:

1. **Search Space Boundaries:** The symbolic discovery engine is strictly confined to predefined basis function libraries. The framework cannot discover functional forms outside of this dictionary.
2. **Heuristic Selection:** The large language model operates strictly as a heuristic selector and orchestrator of predefined workflows; it does not generate open-ended hypotheses.
3. **Single-Seed Evaluation:** All benchmarks in this paper are evaluated on a single-seed basis (Seed 42) to ensure deterministic consistency. Multi-seed statistical variance is not reported.
4. **Computational Cost:** Iterative Neural ODE integration and layer-wise SVD decompositions introduce small but non-negligible computational overhead compared to purely discrete models.

6 Conclusion

We presented a representation-aware symbolic system identification framework. By formalizing a numerically stable EV3 metric and introducing a geometry-driven selection heuristic, our automated pipeline successfully isolates physical laws under dense noise. Future work will extend the framework to partial differential equations and evaluate multi-seed statistics.

References

Steven L Brunton, Joshua L Proctor, and J Nathan Kutz. Discovering governing equations from data by sparse identification of nonlinear dynamical systems. *Proceedings of the National Academy of Sciences*, 113(15):3932–3937, 2016.

Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural network representations revisited. In *International Conference on Machine Learning*, pages 3519–3529, 2019.

Chris Lu et al. The ai scientist: Towards fully automated open-ended scientific discovery. *arXiv preprint arXiv:2308.07332*, 2023.

Ari Morcos, Maithra Raghu, and Bengio Samy. Insights on representational similarity in neural networks with projection weighted cca. In *Advances in Neural Information Processing Systems*, volume 31, 2018.

Christopher Rackauckas, Yingbo Ma, Julius Martensen, Collin Warner, Kirill Zubov, Rohit Supekar, Dominic Skinner, Ali Ramadhan, and Alan Edelman. Universal differential equations for scientific machine learning. *arXiv preprint arXiv:2001.04385*, 2020.

Maithra Raghu, Justin Gilmer, Jason Yosinski, and Jascha Sohl-Dickstein. Svcca: Singular vector canonical correlation analysis for deep learning dynamics and interpretability. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

Silviu-Marian Udrescu and Max Tegmark. Ai feynman: A physics-inspired systematic approach to symbolic regression. *Science Advances*, 6(16): eaay2631, 2020.

A Axiomatic Derivation of Stable EV3

This appendix derives the Stable Effective Volume (EV3) metric. Given an activation matrix $X \in \mathbb{R}^{N\times D}$, we first construct the centered activation matrix $\tilde{X}$:

$$\tilde{X} = X - \frac{1}{N}\mathbf{1}_{N\times N}X \tag{3}$$

The empirical covariance matrix of the activations is:

$$C = \frac{1}{N}\tilde{X}^\top \tilde{X} \tag{4}$$

Through SVD on the centered activations $\tilde{X} = U\Sigma V^\top$, we extract the singular values $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_k \geq 0$, where $k = \min(N, D)$. The eigenvalues of C are $\lambda_i = \sigma_i^2/N$.

The volume of the covariance ellipsoid is proportional to the product of its semi-axes:

$$\text{Vol} \propto \prod_{i=1}^{k} \sigma_i \tag{5}$$

In high dimensions, standard determinants collapse to 0.0 due to floating-point underflow. We define the stable effective volume as the dimension-normalized geometric mean in log-space:

$$\text{EV3}(X) = \exp\left(\frac{1}{M}\sum_{i=1}^{M}\log\sigma_i\right) \tag{6}$$

where M is the count of active singular values exceeding a regularizing threshold $\epsilon = 10^{-10}$.

A.1 Properties and Proofs

1. **Rotation Invariance:** Let $R \in \text{SO}(D)$ be an orthogonal rotation matrix. The rotated activations are $X' = XR$, which yields centered activations $\tilde{X}' = \tilde{X}R$. SVD on $\tilde{X}'$ is $\tilde{X}' = U\Sigma(V^\top R)$. Since the product of two orthogonal matrices $V^\top R$ remains orthogonal, the singular values Σ are unchanged. Thus, $\text{EV3}(XR) = \text{EV3}(X)$.

2. **Non-Negativity:** Since all active singular values satisfy $\sigma_i > \epsilon > 0$, their logarithms are real, and EV3 is strictly positive (EV3 > 0). When the representation collapses to a single point, $M = 0$, yielding EV3 $= 0.0$.

3. **Subspace Stability:** In deep networks, representations often collapse to a lower-dimensional manifold $d < D$, forcing $D - d$ eigenvalues to zero. A standard determinant collapses immediately to 0.0. In contrast, EV3 filters out the collapsed directions ($\sigma_i \leq \epsilon$) and measures the volume spanned strictly within the active subspace.